

LUKE MONINGTON

Data Scientist

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Methodical, analytical, and passionate professional with progressive years of experience in delivering development initiatives through high-level machine learning competencies and project management.

- Proficient in building codes, algorithms, and networks and directing project lifecycles through collaboration, streamlined processes, time management, leadership, and quality strategies.
- Proven capacity to interface at all levels of the organization to achieve productivity, scalability, and maintainability.
- Leverage technical and business intelligence in formulating, developing, and implementing innovative solutions to develop reliable systems and AI technologies and deliver optimum operations.
- Data Scientist who thrives in a fast-paced startup environment and can competently manage progressive projects and programs of increasing complexity while ensuring a positive operational quality and efficiency.

CORE COMPETENCIES:

Training Deep Learning Models on Large Datasets | Directing Data Annotation Teams | Evaluating Human Pose Model Architectures | Cloud Architecture (GCP, AWS, Azure) | Architecture Design Patterns | Natural Language Processing | Data Storage Technologies (SQL & NoSQL) Databases | Maintaining Data Pipelines to Support NLP Workflows | Data Processing | Understanding Business Requirements | Computer Science | Providing Technical Mentorship | Strong Programming Skills | Ability to Develop Solutions that Meet Requirements | Ability to Work Independently | Strong Problem-Solving Skills | Model Deployment | Experimentation & A/B Testing | Model Interpretability & Explainability (SHAP, LIME)

TECHNICAL SPECIALTIES

Platforms: Linux, Windows, Android, iOS

Tools/Software: Jupyter Notebooks, Spyder, Docker, Jira, Git, VSCode, VisualStudio IDE, MySQL, eFront Software Suite, Tableau, Xamarin, .NET Framework, WordPress, Model Context Protocol (MCP), agentic LLM tool design

Frameworks/Libraries: Flask, Pandas, NumPy, SciPy, scikit-learn, matplotlib, XGBoost, TensorFlow, PyTorch, OpenCV, Keras, Semantic Kernel, EF Core, Blazor WASM, MudBlazor, Kernel Memory

Programming Languages: SQL, Python, R, C#, HTML5, CSS3, XML, frontScript, Octave, C/C++, Java

Databases: Oracle SQL

Hardware: Raspberry Pi, Arduino

Distributed Big-Data Computing Solutions: Hadoop, Spark, Hive

Source Control Systems: Git

Container-Ecosystem: Docker, Mesos, Kubernetes

Cloud-Based Development Platforms: AWS, Azure, GCP

PROFESSIONAL EXPERIENCE

MACHINE LEARNING ENGINEER, Artisight | Remote Work

Jun 2022 – Present

Spearhead data science strategies to deliver machine learning, deep learning solutions, and computer vision services.

- Built production model and data drift monitoring on Databricks with Apache Spark and PySpark for millions of packets per minute across dozens of facilities and with live alerts through OpsGenie.
- Designed and executed scale testing from 1x to 300x on Databricks to prepare for a nationwide rollout. Instrumented each ETL pipeline stage, benchmarked pandas versus Spark, and quantified resources needed at various scales.
- Delivered a distributed inventory tracking system across facilities using PySpark for preprocessing and cleaning, XGBoost for classification, and a post processing pipeline to track asset movement in real-time across locations.
- Built a Model Context Protocol (MCP) agent framework that exposed hospital specific tools to LLMs, enabling agentic reasoning and allowing the model to autonomously call MCP tools to answer real time questions from end users.

Major Project

- Led the design and execution of a comprehensive project to replicate and implement an esteemed machine learning research paper on predicting patient outcomes, specifically “Length of Stay” and “Chance of Readmission”.
- Established numerous GPU clusters on AWS for distributed training and fine-tuning of Large Language Models (LLMs).

PROFESSIONAL EXPERIENCE *CONTINUED*

- Pretrained the LLM on a massive dataset comprising hundreds of millions of clinical patient notes, harnessing the power of generative AI to ensure detailed understanding and processing of medical terminologies and contexts.
- Achieved predictive accuracies with Area Under the Curve (AUC) values rivaling those reported in the original research: 0.945 for mortality and 0.881 & 0.889 for short and long ICU stays respectively.

DATA SCIENTIST, KPMG | New York, NY**May 2021 – Jun 2022**

Facilitated a culture of innovation, implemented solutions in a machine learning pipeline, and leveraged best practices while working on multiple large NLP projects to extract information from text using regex, Transformers, DNNs, and XGBoost.

- Implemented end-to-end machine learning solutions, including natural language processing (NLP), regression, and classification models, to extract insights and improve predictive accuracy across large datasets.
- Implemented pipelines for preprocessing, feature engineering, and model training.
- Utilized machine learning expertise in predicting client turnover using classification and data analysis.
- Successfully extracted information from hundreds of thousands of contracts with more than 98% accuracy.
- Achieved 99% accuracy by effectively building out the optical character recognition (OCR) capabilities of the models through experimentation with various solutions, such as different OCR models, OCR hyperparameters, and text rectification techniques (rotation, slant correction).

MACHINE LEARNING ENGINEER, Resource Capital Funds | Denver, CO**May 2020 – May 2021**

Answered business questions by implementing machine learning algorithms on Jupyter Notebooks such as XGBoost and K-Nearest Neighbors.

- Automated 10% of the company's financial opportunity investment decisions by building six backend machine learning models and a front-end GUI, which predicted the locations of gold and other precious metals in the ground.
- Developed and maintained MLOps workflows, including automated model training, deployment, and monitoring pipelines (CI/CD), ensuring scalable and reliable machine learning systems in production.
- Demonstrated solid data science knowledge in deploying a live graphical user interface (GUI) on AWS EC2 Instance.
- Built and deployed an end-to-end ML pipeline for geospatial data analysis, including ETL, feature engineering, regression/classification, and hyperparameter tuning, ensuring production-ready models for mining site prediction.

DATA SCIENTIST INTERN, Spectrum | Centennial, CO**May 2019 – Aug 2019**

Worked in a highly collaborative team environment, drove new technology development, and generated insights into different residential construction project attributes and factors that may affect project success.

- Adroitly performed various statistical analysis techniques and compiled code using Python and R.
- Leveraged expertise in Principal Component Analysis (PCA), Multiple Correspondence Analysis (MCA), Odds Ratio, Chi-Squared, and Regularized Logistic Regression.
- Translated complex statistical analyses into executive-level insights, achieving 5% reduction in project expenses.
- Applied statistical hypothesis testing methods, including t-tests, ANOVA, and Kolmogorov-Smirnov (KS) tests, to evaluate data distributions and validate model assumptions, improving reliability of analytical insights.

EDUCATION***Bachelor of Science in Computer Information Systems***

University of Northern Colorado - Monfort College of Business, May 2020 - *Magna Cum Laude*, GPA: 3.85/4

Affiliations:

President/Founder, Artificial Intelligence Club and IoT Club, 2018

CERTIFICATIONS

Machine Learning, Andrew NG | Jan 2019

Azure AI Engineer AI-102, Microsoft | Feb 2022

Azure Fundamentals AZ-900, Microsoft | Jan 2022